

AI LITERACY FOR SOCIAL SCIENTISTS · SPECIAL LECTURE 01

사회과학자를 위한 AI Literacy 특강 시리즈

Rethinking the Research Process

How AI Is Changing Research for Students and Professors

Steven Denney

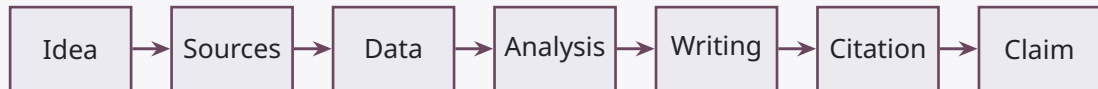
Leiden University

Korea University · 18 September 2026 · Moderator: Sung Eun Kim

Where I am speaking from

- Institute for Area Studies, Leiden University.
- Experimental social science on Korea and East Asia: surveys, conjoint experiments, text as data.
- Two public repositories this talk draws on: **Open Science Skills**, a library of standards for coding agents, and **AI for Research**, its demonstrations and lectures.
- One running example: an NWO project on immigration attitudes, from grant to conference talk.

One pipeline, taught and run



PART 1

Where we are. Why research is the hard case, and three modes of engagement.

PART 2

Building the check. Four controls, three of them on one project.

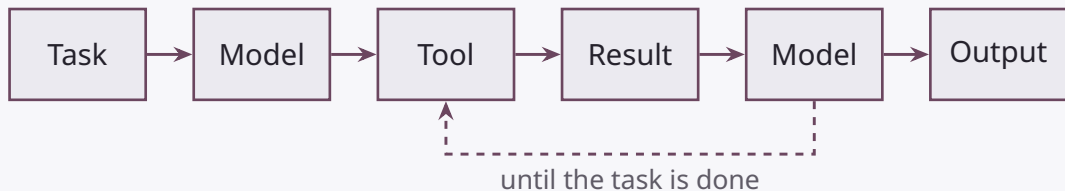
PART 3

One project, start to finish.

PART 1

Where we are

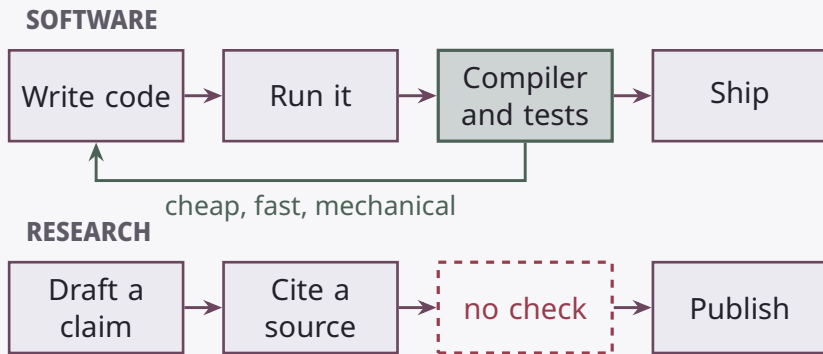
These tools were built for software



A chat window answers. An agent acts, sees what happened, and decides the next step.

The loop after Kenny 2026.

Code runs or it fails. A claim does not.



The mistake surfaces at peer review, in someone else's citation of you, or never.

Three consequences

Errors are silent

And they compound. A bad transcription becomes a bad count, becomes a claim, becomes someone else's citation of your claim.

Checking needs the skill

You cannot tell whether a summary of Brubaker is right unless you have read Brubaker.

The inputs are not in the training data

Your archive. Your unpublished data. A scanned textbook no one has transcribed.

Three modes

AUTOMATE 자동화

Hand off the work

Efficiency is the goal

PROTECT 보호

Preserve the cognitive work

Learning and judgment are the goal

COLLABORATE 협업

Combine human and machine

Reach is the goal

The question is not whether AI can do a task, but what role human cognition should play in doing it.

Each mode has a mechanism

Automate where a wrong answer is cheap or obvious.

Protect where the only check is having the skill yourself.

Collaborate where you can build the check.

Mode 1. Aggressively and shamelessly automate

Cognitive offloading is good.

- Planning and management
- Categorizing email
- Tracking research
- Chores and admin

One month of dictation on two machines: 662 dictations, 54,796 words, about eight dollars in API fees.

Mode 2. Protect. Little to no AI

SURE

- Grading, maybe all of it
- Final manuscripts
- Most teaching
- Reading

NOT SURE. I WILL ASK YOU.

- Writing before the final draft?
- Seed ideas?

Who bears the cost

IN RESEARCH

Your own skill formation.

Yours to waive. You know
what you gave up.

IN TEACHING

A student's formation.

Not yours to waive.

**The student has an interest in the outcome, and only you
can protect it.**

Same stage, different mode

	Student	Researcher
Idea	Protect	Protect
Sources	Protect	Collaborate
Data	Collaborate	Collaborate
Analysis	Protect	Collaborate
Writing	Protect	Protect
Citation	Collaborate	Automate
Claim	Protect	Collaborate

Reading sits in Protect for the student and in Collaborate for the researcher.

Same stage. Different person.

Mode 3. Collaborate

Where you can build the check.

- Research scaffolding and knowledge bases
- Bibliometrics and formatting
- Data collection, organization, analysis, visualization
- Ideation and quality assurance
- Supervision

The rest of this talk is four controls, built.

PART 2

Building the check

One project, one pipeline, four controls

Grant	NWO Open Competition XS, 406.XS.25.01.065. €50,000, October 2025 to September 2026.
Question	Do citizens judge immigration policy by what it does, or also by how governments exercise authority in making it?
Design	Within-subject conjoint, 360 cells, ten tasks and one repeat. United States, Germany, South Korea, Singapore.
Respondents	10,000 after quality checks: Germany 2,502, Korea 2,500, Singapore 2,500, United States 2,498.
Output	Three papers, one conference talk, one pre-registration.
Repository	447 commits, 80 handoff entries, 54 session logs.

A skill is a written standard

```
---  
name: citation-check  
description: Audits a manuscript's citation layer, including  
  in-text and reference-list parity, fabricated or nonexistent  
  sources, DOIs resolving to another work, APA 7 or named-journal  
  style, completeness, and whether each source supports its  
  attached claim. Verifies against Crossref, OpenAlex, DataCite,  
  and Semantic Scholar, audits LaTeX from cited keys, and marks  
  unchecked items as NOT CHECKED.  
---
```

Forty-one of these. The agent loads one when the task matches, or when you name it.

The boring folder structure is load-bearing

```
<repo>/
├── sources/
│   ├── og/           # original PDFs, gitignored
│   ├── md/           # Markdown conversions, tracked
│   ├── unprocessed/  # drop zone for new PDFs
│   ├── references.bib # one entry per source
│   └── README.md      # the convention, written down
├── scripts/convert-sources.sh
├── CLAUDE.md          # conventions the agent reads
└── AGENTS.md -> CLAUDE.md
```

The source library is the spine. The rest grows from it.

As built

The NWO source library, 16 September 2026

Markdown files in sources/md/	119
of which verbatim conversions	103
of which notes for works not held	15
of which an index of those notes	1
Original documents in sources/og/	103
Entries in the master bibliography	126
Files waiting in the drop zone	0

What a converted source looks like

```
sources/md/  
  hainmueller-hopkins-yamamoto-2014-causal-inference-conjoint.md  
  
Political Analysis (2014) 22:1-30    doi:10.1093/pan/mpt024  
  
# Causal Inference in Conjoint Analysis: Understanding  
  Multidimensional Choices via Stated Preference Experiments  
  
Jens Hainmueller    Department of Political Science, MIT  
Daniel J. Hopkins    Department of Government, Georgetown  
Teppei Yamamoto    Department of Political Science, MIT
```

Plain text the model reads directly. That is what makes a grounded check possible.

A note is not a source

The banner on every note for a work the project does not hold

THIS IS NOT A SOURCE. It is a repo-authored note about a work we do not hold. Nothing here may be used to verify a claim the paper attributes to this work. A fact-check must record such a claim as unverifiable.

Paper 2 cites 74 works: 62 with the real text, 12 with a note, none with nothing.

`sources/md/* .NOTE.md`. Knowledge base declared complete 27 August 2026.

The nearest thing to a compiler

- Crossref and OpenAlex: do title, author, and year match an indexed work?
- DOI resolution: does the DOI point at *this* work?
- Exact-title search: zero hits anywhere is a strong signal.
- Author-corpus check: does the title appear in the real author's works?

Never treat “DOI resolves” as “DOI correct.” A live DOI pointing at a different real paper is the classic fabrication signature.

What citation-check can say

- MISSING DOI
- DEAD DOI
- DOI RESOLVES TO DIFFERENT WORK
- METADATA MISMATCH
- TITLE DRIFT
- STATUS UPDATE
- NEEDS AUTHOR VERIFICATION
- LIKELY FABRICATED

The lookups are mechanical: an index either holds the work or it does not. Whatever was not checked is reported as NOT CHECKED.

What it found, June 2026

37 cited works without a source file, checked. All 37 exist.
Eight bibliography entries wrong. One of them invented.

FitzgeraldCurtisCorliss2012	invented: wrong authors, title, journal real work: "Anxious Publics", CPS 45(4)
Solodoch2021PS	dead DOI, fabricated subtitle
CitrinMuste1999	misattributed: Citrin et al. 1997
HarellBantingKymlicka2022	wrong DOI
NewmanHartmanLownFeldman2015	wrong DOI

Five of the eight, from HANDOFF.md, 30 June 2026. All eight corrected the same day.

The gate never calls a model

```
$ check_deliverable.py --fast
PASS manifest      manifest ok: nwo26-p2-apsa-2026-talk (talk)
PASS credentials   no credential patterns in 18 files
FAIL citations     unverified: Avina et al. 2026           content.md:93
FAIL citations     unverified: Avina et al. 2026           content.md:147
FAIL citations     unverified: Ogura et al. 2026           content.md:217
PASS numbers       147 table values present in numbers.tex
                   (snapshot 2026-08-31T1349Z)
WARN prose_tells   em dash, house style is a spaced en dash   x4
--- FAIL 3, INFO 1, PASS 3, WARN 4                               exit 1
```

Three citations the script could not resolve. 147 numbers traced to one snapshot. No model, no judgment, in seconds.

What happened next

Commit to the paper's bibliography, 14 September 2026

Add Ogura et al. 2026 to the bibliography.

The APSA 2026 deck cites Ogura, Miwa, Kobayashi and Kasuya (POQ 2026) on symbolic versus operational ideology in Asian and Western countries. **The deliverable gate checks every slide citation against this file, so the entry lives here even though the paper does not yet cite it.**

The cache today: 23 references verified. The gate changed what the author did, not just what the author knew.

It refuses to run on an empty base

```
# Fact-Check -- Pre-flight: NOT READY
```

```
Reason: coverage below the bar
```

```
Knowledge base searched: sources/md/
```

```
Cited works with no Markdown file: <keys>
```

```
## To proceed
```

```
1. Convert and file the missing sources
```

```
2. Confirm each cited work has sources/md/<author>-<year>-<slug>.md
```

```
3. Re-run /fact-check
```

Below two-thirds coverage, it stops. A check run around a hole is worse than no check.

What fact-check can say

- SUPPORTED
- PARTIALLY SUPPORTED
- UNSUPPORTED
- CONTRADICTED
- MISATTRIBUTED
- SOURCE INSUFFICIENT
- NOT IN KB

A model reads the file, not its memory. A verdict of support or contradiction carries a verbatim quote from the source. A missing source is reported as missing, not judged.

What it found, August 2026

Every claim in Paper 2 checked against the source text where we hold it, not the citation. Nineteen corrections. Four citations had supported the opposite of what their source found.

Cited work

What it actually says

Hangartner et al.

The study is Greece, not Germany.

Tichenor

Immigration cut *across* party lines.

Hooghe and Marks

Nothing on asylum or migration governance.

Citrin et al. 1997

A null on personal economic competition.

One finding in full

The claim	An AMCE, the average effect of one attribute in a conjoint, is defined over the levels a researcher writes.
The source	Bansak, Hainmueller, Hopkins and Yamamoto: an AMCE is always defined with respect to a baseline value and to the randomization distribution. Zero hits in the chapter for <i>wording, comparable, or cross-attribute</i> .
The fix	The sentence was rewritten to what the chapter says.

This was the last unverified citation in the paper.

Two checks, two different failures

CITATION-CHECK

ferreira-nair2021

Well formed, consistent with the text. ✓ **Passes**

The format check has nothing to object to.

FACT-CHECK

“compulsory voting also raises citizens’ political knowledge”

✗ Fails CONTRADICTED

Source: “Whatever compulsory voting does, it does not appear to teach.”

Formatting-clean is not the same as true.

The public sample project in the AI for Research demos, with planted errors.

The gate reads what it is told to read

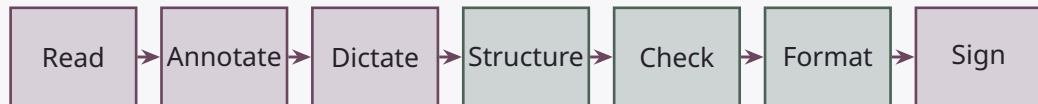
Lint finding F011, 9 September 2026, on a deck already at a public URL

This citation is inside a text field while its sibling cite field is left empty, so the citations check, which reads the cite fields, never sees it.

“Ogura” has no entry in the bibliography, so **an unresolvable citation stands on a published slide, and the gate cannot catch it.**

The first run that morning caught two of three. The model found the third. The scan was widened the same day, and the second run caught all three.

The same logic runs the supervision desk



 the supervisor

 the machine

The judgment stays with the supervisor. The machinery around the judgment is automated.

Dictation, before and after

RAW TRANSCRIPT

“Okay, I have a number of things that I want us to address, some of which will **beckon you to engage** with the open science skills that we’re using here to update them.”

AFTER CLEANUP, ABOUT A SECOND LATER

“Okay, I have a number of things that I want us to address, some of which will **require you to engage** with the open science skills that we’re using here to update them.”

One repair. Nothing rewritten.

The dictation log, 4 June 2026.

Structure and check

STRUCTURE

Five readers in parallel, one per assessment criterion. Each writes findings with a page anchor.

CHECK

One cross-checker verifies that every quoted span appears at the cited page, and drops what it cannot find.

The rule in every reader's instructions

Quote the span verbatim with its page anchor. If you cannot quote it, do not assert it.

A hard cap of 900 words. The letter is pasted into the faculty form by hand. One criterion, the student's learning process, is never marked by the machine.

The same letter, a third shorter

FIRST DRAFT, 89 WORDS

This has a compelling premise for a book: a persuasive case for immigration requires honesty about distributional costs and attention to policy design. The insider perspective gives the argument stakes, and the “laundering pipeline” is a memorable way to explain how qualifications disappear between research and public communication. The Germany example is especially useful because it distinguishes disproportionate representation among suspects from changes in local crime. The discussion of migrants’ agency and gains from mobility gives the chapter a constructive destination. Those elements could anchor an accessible, consequential book.

AFTER THE EDITING PASS, 61 WORDS

The chapter has a strong premise. Immigration outcomes depend on who moves, under which rules, and over what period. The insider account adds weight, and the “laundering pipeline” is memorable. Its distinction between suspect overrepresentation and changes in total crime is useful. It argues for policies that respect migrants’ agency. The book is worth pursuing, but the chapter needs stronger evidence.

**Automate the machinery, not
the judgment.**

PART 3

One project, start to finish

Twelve months, eight stages

When	Stage	The machine	Stayed human
2025 Mar to Jun	Grant and ethics Design and plan	no use on record adversarial reviews, bibliography audits	everything on record every design decision, logged
Mar to Aug	Instrument	survey from a text spec, blind translation reviews	native-speaker checks
10 Aug	Pre-registration	documents generated from the LaTeX source	the deposit itself
11 to 31 Aug	Fielding	quota grids, publish gating, second reads	vendor letters, quota rulings
31 Aug	Analysis	figures, numbers . tex, verdict guard, the runs	estimand and rule, sign withheld
Aug to Sep	Writing	prose passes, pre-submission review	the wording, and every push
4 to 9 Sep	Talk and audit	deck build, then gate and lint	the notes, the order, the fixes

What stayed human, by rule

notes/agentive-analysis-integrity.md

Confirmatory analysis is blind to the hypothesized sign. Give the agent the estimand and the decision rule, not the direction we hope for.

CLAUDE.md

Do not change a paper's confirmatory hypothesis structure without discussion.

.claude/skills/finished/SKILL.md

Never push to Overleaf without the PI's say-so.

Every number resolves to one snapshot

```
% GENERATED FILE -- DO NOT EDIT.  
% Written by tools/paper-figures/emit_numbers.py; hand edits are lost  
% on the next data refresh and break the numbers-discipline audit trail.  
%  
% Paper:                Paper 2 (Experiment 2)  
% Snapshot of record: 2026-08-31T1349Z  
% Keys defined:         2018  
%  
% Every result literal in this paper resolves through \rr{<key>}  
% against the definitions below.
```

The verdict guard reports any decision-level key that changed between snapshots. Paper 2: 0 of 107 moved.

Header abridged.

Disclosure

Use of AI, from the manuscript

Generative and agentic AI tools assisted with producing and checking analysis scripts, generating tables and figures, and research ideation, including adversarial reviews and suggestions for interpretation and analysis. The systems used were Anthropic's Claude Opus 4.8/5 and Fable 5 and OpenAI's GPT-5.5/5.6. **All design decisions, hypotheses, estimands, analysis, and interpretations are the author's, who carefully reviewed all input and takes full responsibility for the content of this manuscript.**

What it took

447

commits

80

handoff entries

54

session logs

11,772

words, against a
ceiling of 10,000

The machinery was automated. The paper is still being written.

For students, for professors

STUDENTS

Automate the clerical.

Protect reading, first ideas, and the procedures you are still learning.

Collaborate only where you can check the output.

PROFESSORS

Automate more of your own machinery, and leave a trace.

Protect the formation of your students' judgment.

Collaborate where the machine changes what is feasible.

The question is not whether AI can do a task, but what role human cognition should play in doing it.

Everything shown today is public

OPEN SCIENCE SKILLS

41 skills for Claude Code, 40 for Codex.

github.com/scdenney/open-science-skills

AI FOR RESEARCH

Setup, demonstrations, lectures, this deck.

scdenney.github.io/ai-for-research

CONTACT

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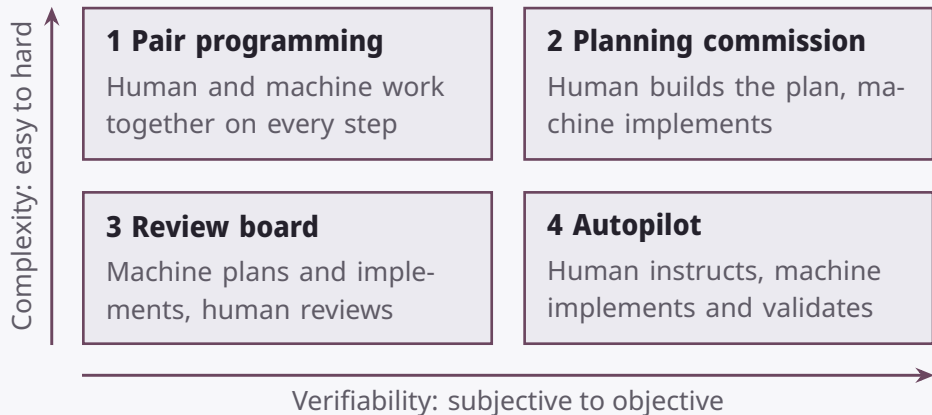


These slides

APPENDIX

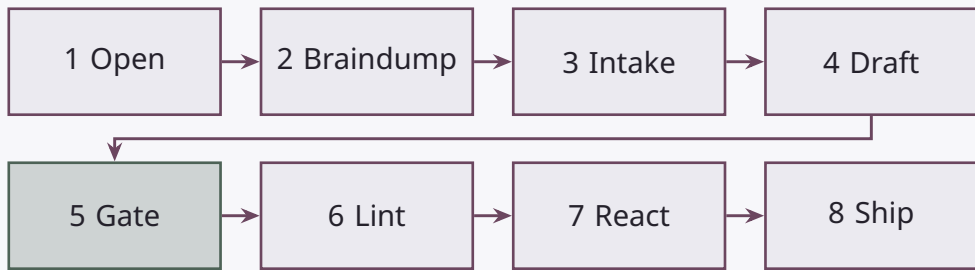
For the discussion

Four working modes, by complexity and verifiability



Kenny 2026, adapting Swaney's 2026 DDSS workshop.

The deliverable pipeline in eight steps



One wiki per piece of research, one manifest per deliverable. The gate is a script that runs at every commit. Lint is a model review that runs only after the gate passes.

Which converter, for which document

Document	Tool	Why
Born-digital PDF, prose	opendataloader-pdf	keeps paragraph boundaries, does not invent tables
PDF whose payload is tables	anydoc	real tables come out as real columns
Office and e-book formats	anydoc	one local binary, fourteen formats
Scanned or image-only PDF	stop, hand to OCR	neither text extractor can read an image
One value out of a clean PDF	pdftotext	fast, no dependencies

A pre-submission review, as run

Method line, Paper 2 review of 30 August 2026

Nine dimension agents, two cross-checkers, one orchestrator. Every quote in the nine finder files was located verbatim in the current source. Zero findings dropped as hallucinated quotes. Eight finder claims dropped or corrected as errors of inference. Nothing in the manuscript, bibliography, or pipeline was edited.

Issues: 9 critical, 40 recommended, about 70 minor. Seven of the thirteen criticals fixed two days earlier were open again, because a consolidation had deleted the section that closed them.

References

Kenny, Christopher T. 2026. "Agentic AI for Political Science Research." Center for the Study of Democratic Politics, Princeton University, 10 September.
github.com/christopherkenny/skills

Swaney, Colin. 2026. "Claude Overload." Data-Driven Social Science workshop, Princeton University, 20 March. github.com/princeton-ddss/claude-overload

Karpathy, Andrej. 2026. "LLM Wiki." April.
gist.github.com/karpathy/442a6bf555914893e9891c11519de94f

Bansak, Kirk, Jens Hainmueller, Daniel J. Hopkins, and Teppei Yamamoto. 2021. "Conjoint Survey Experiments." In *Advances in Experimental Political Science*. Cambridge University Press.

Hainmueller, Jens, Daniel J. Hopkins, and Teppei Yamamoto. 2014. "Causal Inference in Conjoint Analysis." *Political Analysis* 22(1): 1–30.

Denney, Steven. 2026. "Governing Immigration by the Rules." APSA Annual Meeting, Boston, 4 September. NWO 406.XS.25.01.065.